# DHCP in Windows Server Theory

## Task 1: On your Windows Server VM, open the DHCP Manager and create a new DHCP scope named 'CampusWiFi' with an IP address range of 192.168.50.100 to 192.168.50.200 and a subnet mask of 255.255.255.0.

**Procedure & Verification:**

1. Open the DHCP Management Console on Server01.lab.com.
2. Right-click on IPv4 and select 'New Scope...' to launch the New Scope Wizard.
3. On the Scope Name page, enter 'CampusWiFi' in the Name field.
4. On the IP Address Range page, specify:
   - Start IP address: 192.168.50.100
   - End IP address: 192.168.50.200
   - Length: 24
   - Subnet mask: 255.255.255.0
5. Complete the wizard to create and activate the scope.

*Figure 1.1: Opening New Scope Wizard in DHCP Manager*

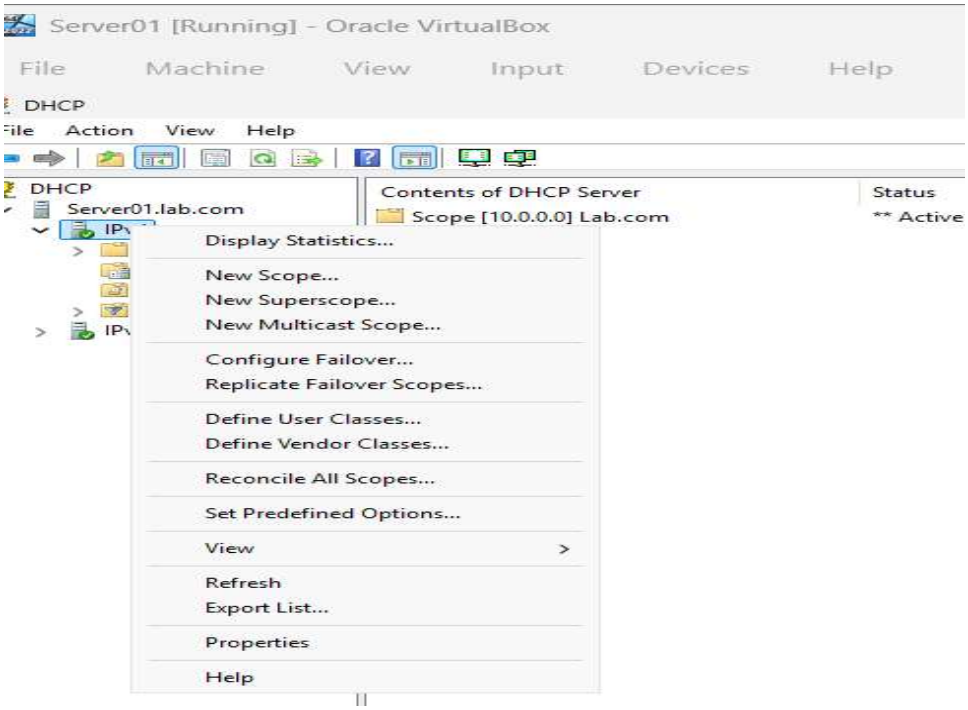
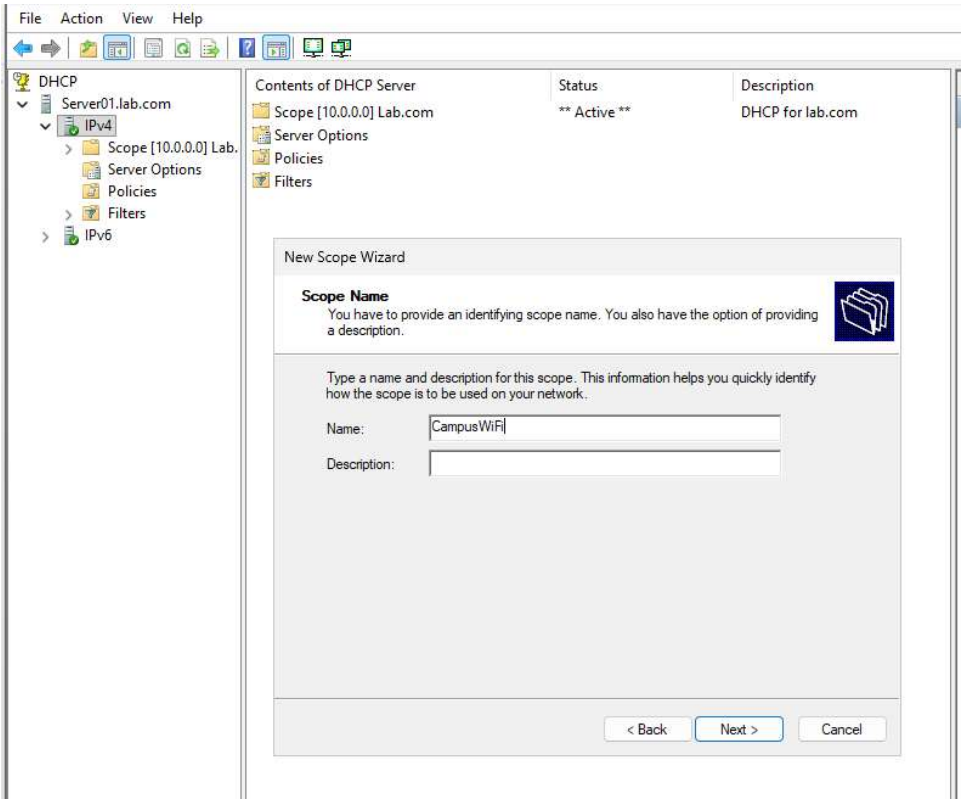


*Figure 1.2: Specifying Scope Name 'CampusWiFi'*

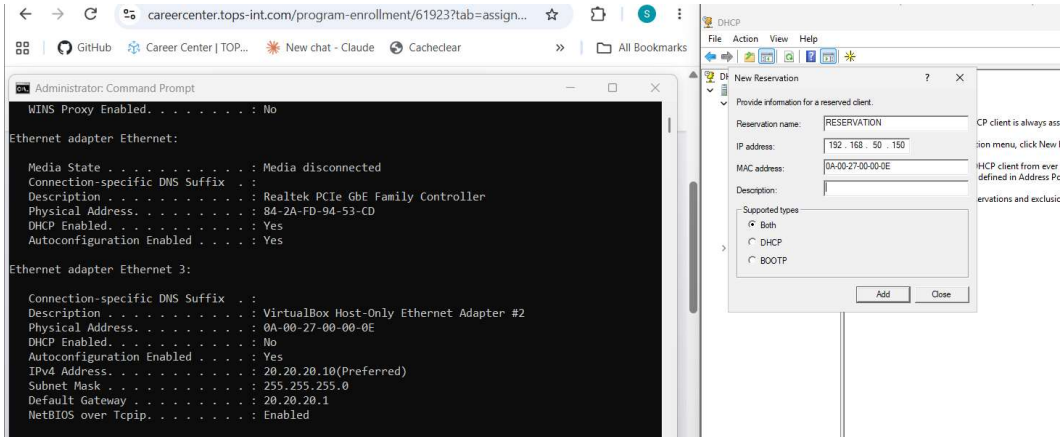



## Task 2: Add a DHCP reservation for a device (use your own laptop's MAC address if possible) so it always receives the IP address 192.168.50.150 from the 'CampusWiFi' scope.

**Procedure & Verification:**

1. Obtain the client device physical address (MAC address). In Command Prompt, running 'ipconfig /all' identifies the physical address of the client network adapter (e.g., 0A-00-27-00-00-0E for VirtualBox Host-Only Ethernet Adapter #2).
2. Expand the newly created 'CampusWiFi' scope tree in DHCP Manager and right-click 'Reservations', then select 'New Reservation...'.
3. Configure the reservation parameters:
 • Reservation name: RESERVATION
 • IP address: 192.168.50.150
 • MAC address: 0A-00-27-00-00-0E
 • Supported types: Both (DHCP and BOOTP)
4. Click 'Add' to store the reservation.

*Figure 2.1: Configuring New DHCP Reservation for MAC Address 0A-00-27-00-00-0E*

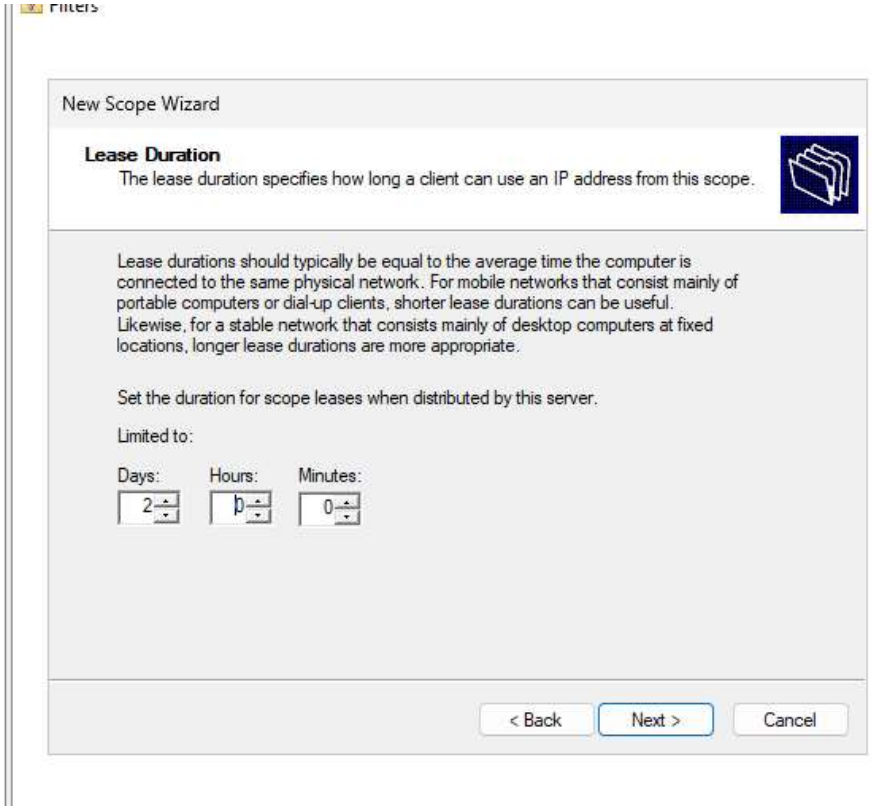



## Task 3: Configure the DHCP lease duration for the 'CampusWiFi' scope to 2 days instead of the default setting.

**Procedure & Verification:**

1. During the New Scope Wizard (or via Scope Properties after creation), navigate to the 'Lease Duration' step.
2. Modify the 'Limited to:' duration fields from the default (8 days) to:
   - Days: 2
   - Hours: 0
   - Minutes: 0
3. Click 'Next' to apply the 2-day lease policy across all dynamic allocations in the scope.

*Figure 3.1: Setting Lease Duration to 2 Days*

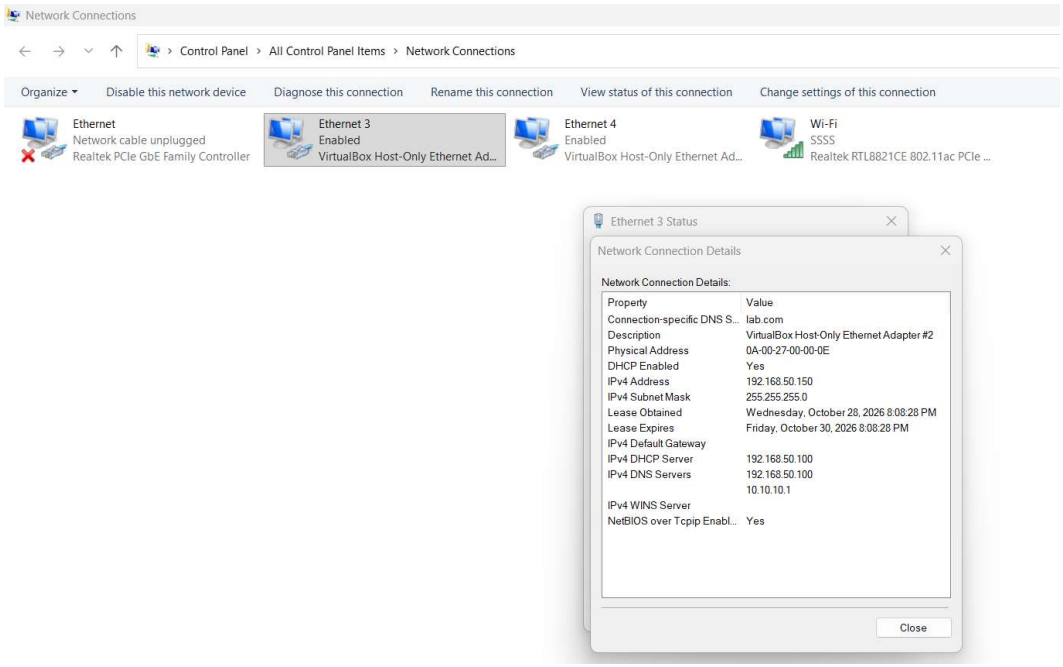



## Task 4: Connect a client machine (or use a virtual client) to the network and verify it receives an IP address from your DHCP scope; take a screenshot showing the assigned IP in the client's network settings.

**Procedure & Verification:**

1. Connect the client network adapter (VirtualBox Host-Only Ethernet Adapter #2) to the DHCP network.
2. Open Network Connections -> Right-click Ethernet 3 -> Select 'Status' -> Click 'Details...'.
3. Confirm client network details:
   - Connection-specific DNS Suffix: lab.com
   - Physical Address: 0A-00-27-00-00-0E
   - DHCP Enabled: Yes
   - IPv4 Address: 192.168.50.150 (Matching the active DHCP reservation)
   - IPv4 Subnet Mask: 255.255.255.0
   - Lease Obtained: Wednesday, October 28, 2026 8:08:28 PM
   - Lease Expires: Friday, October 30, 2026 8:08:28 PM (Exactly 2 days lease duration)
   - IPv4 DHCP Server: 192.168.50.100

*Figure 4.1: Network Connection Details Confirming IP 192.168.50.150 and 2-Day Lease*




**Task 5:** Document your entire DHCP setup process in a Word or PDF file, including screenshots of the scope creation, reservation, lease duration change, and the client receiving its IP address.

This document compiles all five tasks detailing the scope creation, lease duration adjustment to 2 days, MAC address reservation setup, and successful client lease verification with embedded screenshots at each corresponding step.